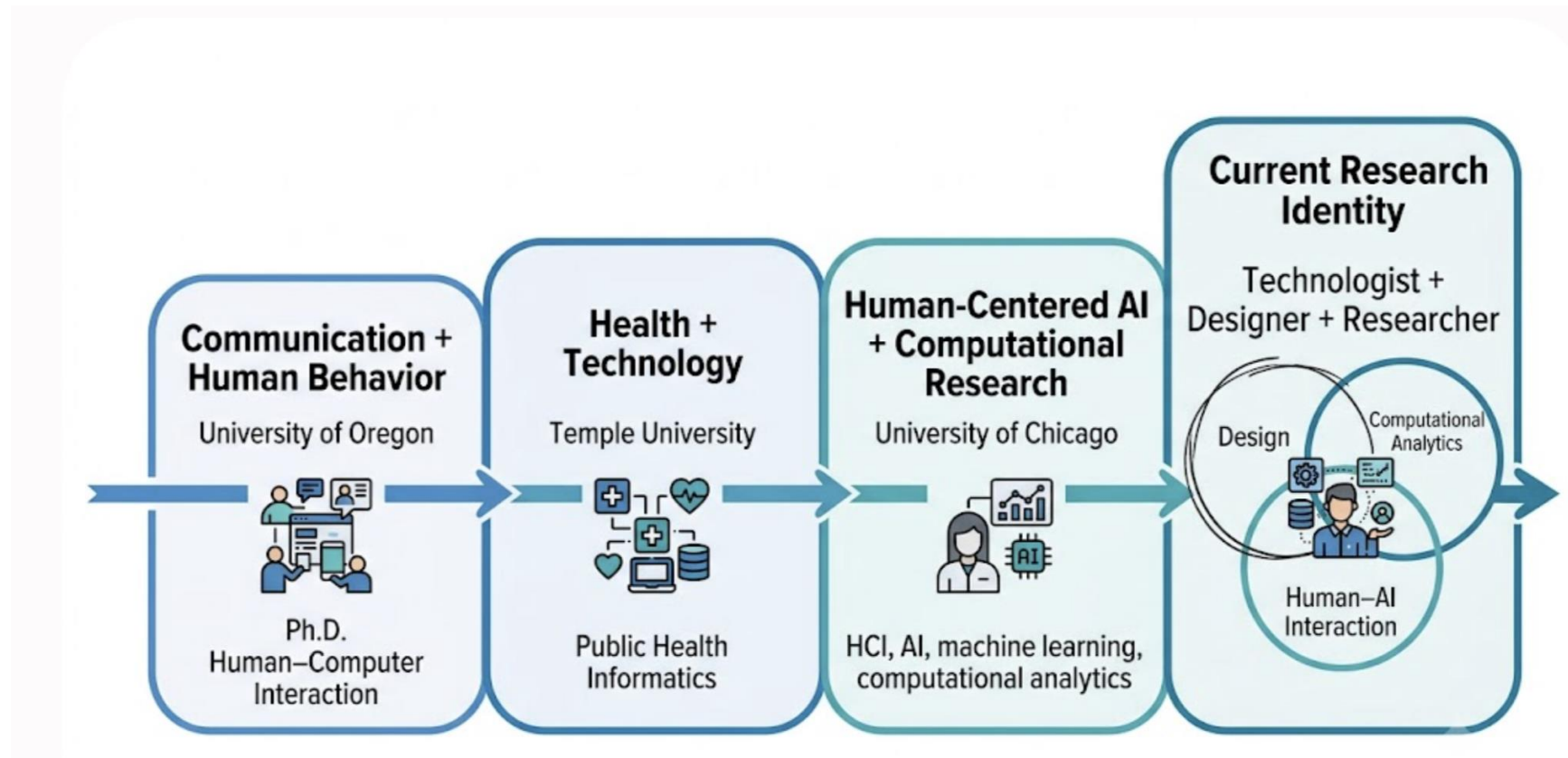


JOUR 701

Research Methods in the Age of AI

My Background

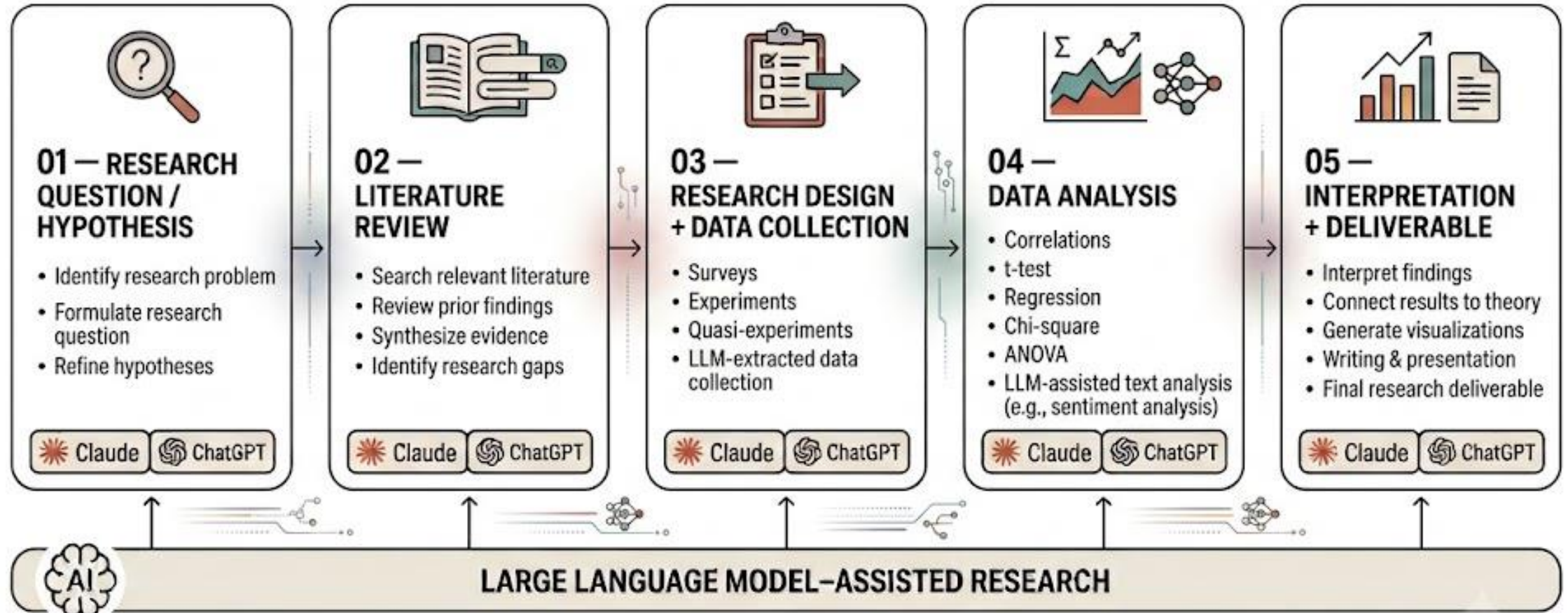
- Technologist • Designer • Researcher



Questions for You

- What do you want to be able to do by December?
- What are you most curious—or nervous—about?

Research Methods in the Age of AI



The researcher remains responsible throughout the entire process.

 + 
GOOGLE COLAB + PYTHON

Research Methods Are More Than Statistical Tests

- Research methods begin before data analysis.
- Tools cannot rescue a weak research question or poor measurement.
- The same tool can be appropriate for one question and inappropriate for another.
- Modern research requires both methodological judgment and technical skill.

How the Course Will Work

- Concept days focus on research logic and interpretation.
- Laboratory days (via ZOOM) turn concepts into practice.
- Google Colab will be our primary computational environment.
- Python will support data management, statistics, text, and AI.
- AI can assist with coding, but you must understand and verify what they run.
- Course materials and official documents will be available through Blackboard and the course [GitHub](#).

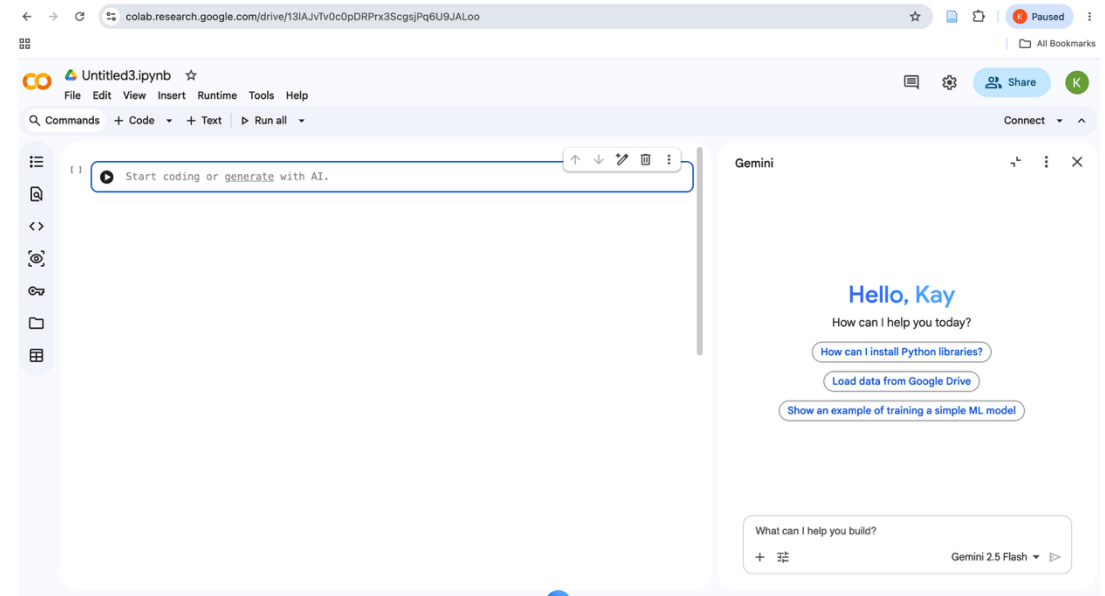
Why Learn Modern Research Tools?

- Research increasingly involves larger and more complex data.
- Employers INCREASINGLY expect quantitative AND computational literacy.
- AI is changing how research is conducted across academia and industry.
- Understanding the tools gives researchers greater control over their work.

The screenshot shows a LinkedIn job search interface. At the top, the LinkedIn logo is followed by a search bar containing 'user experience'. Below the search bar are several filter buttons: 'Jobs', 'Date posted', 'Remote', 'AR/VR', 'IT Services', 'Design', and 'Research & Developm'. Further down, there are more filters: 'Employment type', 'Company', 'Under 10 applicants', and 'In my network'. The search results show '99+ results' for 'United States'. The first three job listings are visible: 1. 'Human-Centered Design (HCD) Expert' by CGI in Fairfax, VA, with 1 benefit and 2 company alumni. 2. 'Scientist - Human Computer Interaction (UX/HCI/HMI)' by Charles River Analytics in Cambridge, MA, with Dental and +2 benefits. 3. 'Design Insights & Validation Research Lead' by HP in Vancouver, WA (On-site), with Medical, Vision, Dental, and +1 benefit. To the right of these listings, a detailed view of the 'Human-Centered Design (HCD) Expert' job is shown, including the CGI logo, job title, location, and a 'Reposted 3 hours ago' status. It also features a 'Full-time' filter, 'Apply' and 'Save' buttons, and a section titled 'Your profile and resun qualifications well' with a 'Show match details' button. At the bottom right, there is a section titled 'People you can reach out'.

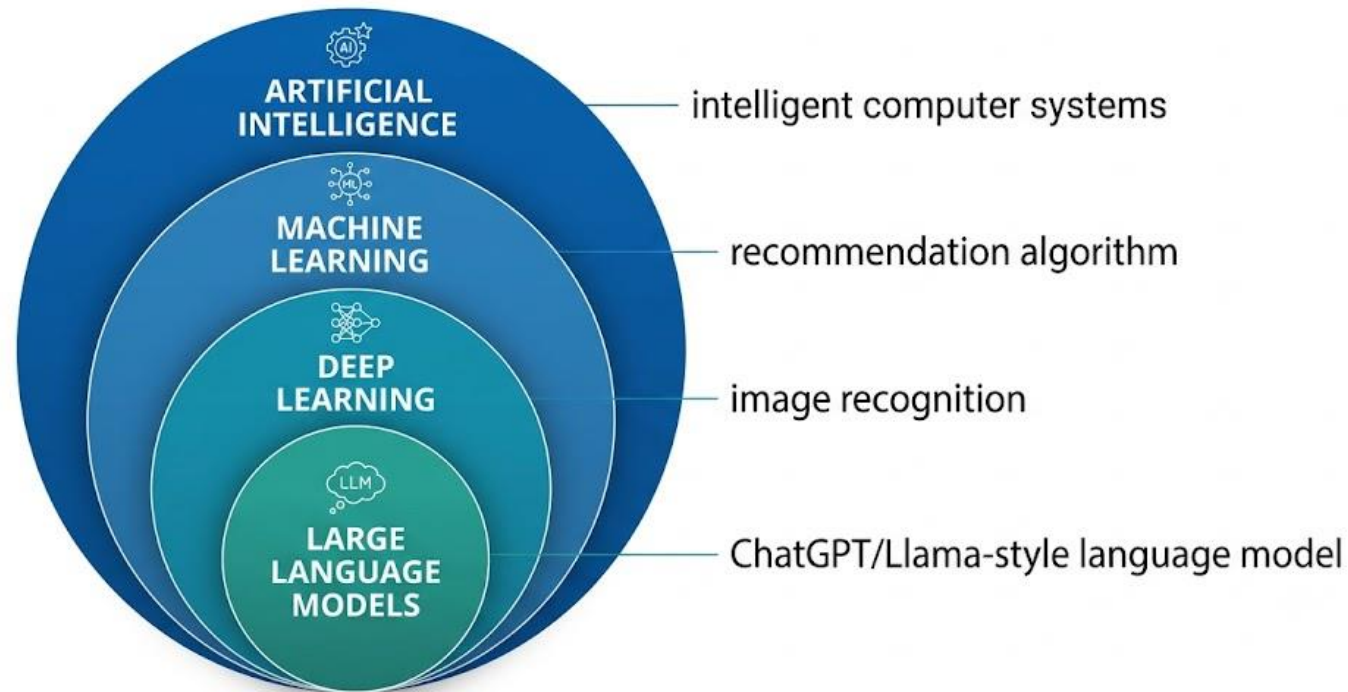
Modern Research Tool: Google Colab

- Colab is a browser-based notebook for running Python.
- Colab makes computational work easier to share and reproduce.
- Computing runs on Google's cloud infrastructure.
- CPU is enough for most course tasks; GPU may help with some AI models.



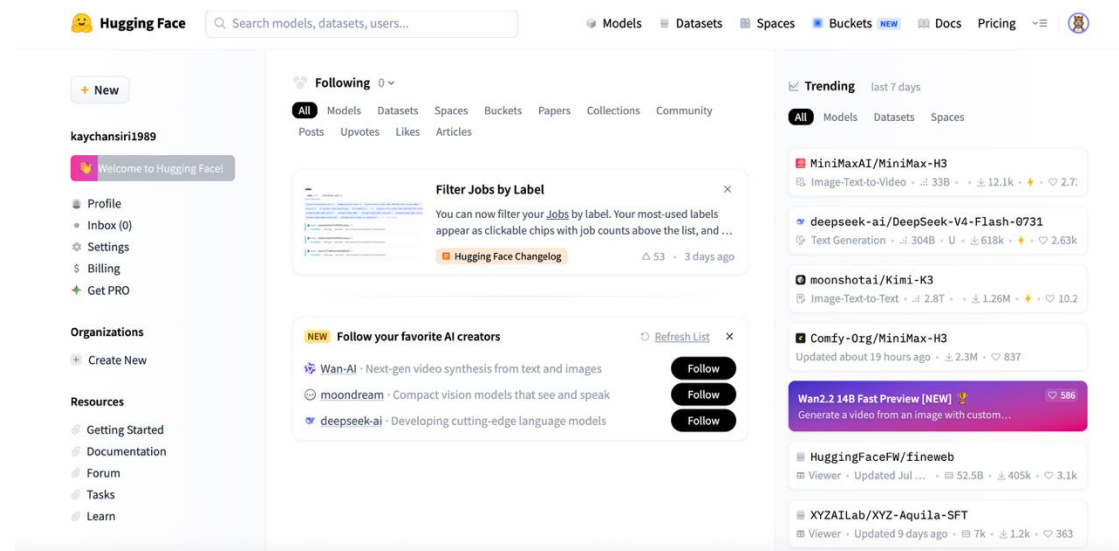
Modern Research Tool: Artificial Intelligence, Specifically LLMs

CONCEPTUAL AI HIERARCHY



Modern Research Tool: Hugging Face

- Hugging Face hosts LLMs, datasets, code, and documentation.
- Researchers can discover and use LLMs from many organizations.
- Many models provide openly accessible weights (AKA structure), while licenses and access conditions vary.
- Proprietary LLMs such as GPT and Claude do not generally release their full model weights.



The Researcher Remains in the Loop

- LLMs can generate suggestions.
- Python can automate procedures.
- Models can identify patterns.
- NONE determines whether the research question is meaningful.
- NONE determines whether the evidence is valid.
- Those remain YOUR decisions.

What I Do Not Expect

- You do NOT need previous programming experience.
- You do NOT need to become an AI engineer.
- You do NOT need to memorize Python syntax.
- You DO need to ask questions, experiment, and verify results.
- Confusion and debugging are NORMAL parts of computational research!

Recap

- Research is a process, not a single technique.
- Modern research combines traditional design with computational tools.
- LLMs expand what we can do with language analysis and the whole research process.
- Python and Colab help make research reproducible.
- Good research still depends on design, validation, and judgment.

Next session: How Do
LLMs Actually Work?